Fiscal Stimulus and Productivity: simulating the NGEU program with an endogenous growth model

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Abstract

This paper introduces an endogenous growth general equilibrium model of firm dynamics and innovative investment for the Spanish economy that allows a better understanding of the medium-term effects of economic policies and shocks. We calibrate the model using both aggregate and firm-level data. We then use the model to assess the macroeconomic consequences of the different components of the Next Generation EU (NGEU) program, including public investment, private capital transfers, and innovative investment transfers. According to our baseline simulation, the NGEU funds significantly foster economic activity, with annual GDP growth increasing between 0.08 and 0.13 percentage points over the period of NGEU funds disbursement. In particular, we find that one of the key drivers of these output gains is the endogenous response of productivity to the fiscal stimulus. Among the different policy instruments, we find that innovation transfers deliver the largest effects on aggregate output, only matched by highly efficient public investment.

Keywords: productivity, public investment, endogenous growth, Next Generation EU

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1 Introduction

At least since Lucas (1987, 2003), it is well understood that economic policies and macroeconomic disturbances that have long-lasting effects on economic activity are of first-order importance for welfare. More recently, the last financial crisis together with the recent pandemic and energy crises have revived the interest in the persistent effects of shocks and subsequent policy responses.[1] A prominent example is the Next Generation EU (NGEU) program enacted by the European Commission in 2020, one of the largest fiscal stimulus measures in the history of the European Union, aimed at fostering a long-lasting and structural recovery of member states after the pandemic crisis (European Council, 2020).

In this paper, we introduce a dynamic general equilibrium model of endogenous growth based on Atkeson and Burstein (2019) that allows us to explore the long-term effects of economic policies. We first extend the model along several dimensions, such as elastic labor supply and productivity-enhancing public capital, and we calibrate it to the Spanish economy using aggregate and firm-level data. We then use the model to study the aggregate consequences of three different broad components of the NGEU funds: public investment, private capital transfers, and innovation transfers. According to our results, the funds associated with the NGEU program can have significant effects on output. In particular, we find that a key driver of the positive effects of the fiscal stimulus is the endogenous response of productivity, driven by the increase in firms' expenditures on innovative investment. In addition, we show that transfers associated with innovation, by fostering productivity, tend to deliver the largest effects on output.

[1]For example, Aikman et al. (2022) discusses the lasting effects on output of deep contractions, Jordà et al. (2022) documents the persistent economic consequences of pandemics, Anzoategui et al. (2019) provides a model where weak aggregate demand leads to a slowdown of productivity growth, Garcia-Macia (2017) shows that the interaction between financial frictions and intangible investment generates persistent slumps, and Schmitz (2021) shows that firm heterogeneity amplifies the effects of financial crises on aggregate innovation in a model calibrated to the Spanish economy.

More in detail, we consider a real model of firm dynamics of a closed economy – with the exception of international transfers. Firms use physical capital and labor as production inputs. These are supplied elastically by households. In addition, firms engage in innovative investment: incumbent firms can invest to create new products (Romer, 1990) or improve the productivity of their own existing products (Aghion and Howitt, 1992), while entering firms carry out innovative investment that leads to the creation of new products. This feature sets the current paper apart from previous papers analyzing the effects of the NGEU program (e.g. Pfeiffer et al. 2022; Fernández-Cerezo et al. 2023) – and more generally the effects of public investment (Leeper et al., 2010; Ramey, 2020; Peri et al., 2023) –, as it allows us to take into account that aggregate productivity is endogenous to firms' innovative investment decisions, and hence to economic policy.

We calibrate the model to the Spanish economy using aggregate data from 2000 to 2019 and firm-level data from the Central Balance Sheet Data Office, maintained by *Banco de España*, to discipline the parameters driving the innovation process of firms. We also show that the predictions of our model are in line with the *R&D* data from *Encuesta sobre Innovación en las empresas*, a firm-level survey conducted by the Spanish Statistical Office (*INE*).

We jointly consider three different policy instruments in this framework, following the breakdown of NGEU projects in Spain elaborated in Fernández-Cerezo et al. (2023). First, we introduce productivity-enhancing public physical capital, which allows firms to use their production inputs more efficiently (Baxter and King, 1993; Ramey, 2020). Since our model distinguishes between investment in private physical capital and investment in innovation, we can model two types of fiscal transfers: one that is used to build physical capital, and one that is used to finance innovative investment. This is particularly relevant, since precisely one of the key pillars of the NGEU program consists of fostering R&D and innovation.[2] In line with the funding design of the NGEU program, we assume that these three policies are initially financed through international transfers that are later paid back, partially or totally, by the fiscal authority through a combination of labor income and consumption taxes. Our simulation assumes that the NGEU funds received by Spain are spent uniformly over time and split between the three different policy instruments in the proportions documented in Fernández-Cerezo et al. (2023).

Our quantitative findings are as follows. First, under our baseline simulation output growth increases by up to 0.13 percentage points per year over the duration of the NGEU program while output increases by up to 1.3% by 2031 (ten years after the start of the program), relative to the scenario without NGEU. Importantly, we find that about 0.5 percentage points of this output expansion is accounted for by the endogenous response of productivity, driven by the increase in innovation investment expenditures. That is, without these productivity gains the size of economic stimulus would be only about 62% of our baseline simulation.[3]

Next, we use the model as a laboratory to evaluate the effects of alternative allocations of funds. This allows us to investigate the relative effectiveness of different policy instruments in terms of fostering economic activity. We consider three different scenarios; in each of them all funds are allocated to just one of the policy instruments considered. We find that innovation transfers tend to deliver the largest effects on aggregate output, comparable only to the effects of public investment if we assume that public capital is highly efficient. Instead, allocating the funds to private capital transfers delivers smaller effects. The reason behind the more limited effects of private capital transfers is that these, by increasing the stock of capital, they lead to a fall in the return of private cap-

[2] See, for example, https://next-generation-eu.europa.eu/index_en.

[3] As we discuss in the main text, the elasticity of aggregate productivity to innovation expenditures predicted by our baseline simulation is in line with the available empirical evidence (Hall, 2011; Moran and Queralto, 2018). Indeed, we show that increasing this elasticity would substantially magnify the macroeconomic effects of the stimulus.

ital. This discourages households' savings, crowding out the accumulation of physical capital financed with private resources. Therefore, physical capital increases by less than one-to-one with transfers. A similar logic also applies to innovation transfers, but this policy still delivers higher effects because innovations that increase aggregate productivity increase aggregate output one-to-one, while the elasticity of output to either private or public capital is smaller than one.

Finally, we compute the fiscal multipliers associated with the baseline simulation of the NGEU program and with each of the alternative scenarios considered. First, we focus on cumulative multipliers after eight years, at the time when the NGEU program expires. In this case, we obtain that the cumulative fiscal multiplier in the baseline simulation is around 0.63. The fiscal multiplier associated with public investment ranges between 0.42 and 1.05, depending on the assumed efficiency of public capital. On the contrary, the fiscal multiplier associated with innovation transfers is consistently around 1.1, while the fiscal transfers to private capital deliver the lowest multiplier. Second, looking at long-run fiscal multipliers we find values that are consistently above one, in particular when the efficiency of public capital is high. Yet, reassuringly, we also find that in the long-run the ranking of fiscal multipliers associated with each alternative allocation of funds holds robustly, with transfers to innovation delivering the highest multipliers unless the public stock of capital built is highly efficient.

Related Literature. This paper contributes to several strands of the literature at the intersection of macroeconomic policies and their persistent effects on economic activity.

First, a few papers have recently investigated the aggregate consequences of the NGEU program. Pfeiffer et al. (2022) studies the effects of the NGEU funds in a New Keynesian multi-country model, finding that cross-country spillovers are an important source of amplification. This paper, however, only considers public investment and abstracts from fiscal transfers. Fernández-Cerezo et al. (2023) introduces a static production network model calibrated to the Spanish economy to analyze the aggregate and sector-level effects of the NGEU funds. The endogenous growth dimension of our model is a distinguishing feature of our paper, allowing us to (i) separately model capital and innovation transfers and (ii) account for the endogenous response of productivity to the fiscal stimulus. As we show in our main simulations, both dimensions are central to understanding the macroeconomic effects of the NGEU program.

Considering firms' innovative investment relates our work to Atkeson and Burstein (2019) or Bloom et al. (2002), who explore the effects on productivity of R&D subsidies. In a similar vein, Akcigit et al. (2021) shows empirically, using cross-sectional data, that changes in corporate taxes tend to affect the innovation decisions of firms. Complementarily, Cloyne et al. (2022) relies on aggregate data to show that corporate tax cuts boost R&D investment, leading to long-lasting effects of this policy. On the other hand, Antolin-Diaz and Surico (2022) documents that military spending shifts public spending towards R&D, boosting aggregate output for longer than public consumption expenditures. In the context of the Spanish economy, Garcia-Macia (2017) shows that transfers to young firms, which tend to be financially constrained, could have boosted intangible investment during the Great Recession, speeding up the recovery. Schmitz (2021), also in a model calibrated to Spain, shows that the presence of firm heterogeneity amplifies the effects of financial crises on aggregate innovative investment. Our work complements these papers by instead focusing on the joint role of fiscal transfers and public investment in boosting output and productivity over the medium term.

Finally, our work also adds to the literature exploring the aggregate effects of public investment. Leeper et al. (2010) emphasizes the role played by implementation delays in a neoclassical growth model. Peri et al. (2023) show that public investment multipliers are amplified in a production network model with sticky prices. Ramey (2020) surveys the literature and explores the mechanisms underlying the propagation of public investment. Our paper complements this literature by also allowing productivity-enhancing capital to spill over to the innovation investment decisions of firms.

Outline. The remainder of the paper is structured as follows. We describe the model and related policies in section 2. Section 3 presents our calibration strategy. We present the results of the simulation exercises, including details of our mapping from the NGEU program to the model, in section 4 . A final section concludes.

2 Model

A fundamental goal of the NGEU funds is to foster economic growth over the medium and long term. In order to do so, funds have been allocated, for example, to public infrastructure and to transfers to firms designed to promote investment in physical capital and improve productivity (Fernández-Cerezo et al., 2023). Therefore, analyzing the macroeconomic consequences of the NGEU program requires a model that is able to capture the medium-term effects of economic policies and that allows for endogenous movements in productivity. Towards this end, we consider and extend the endogenous growth model of Atkeson and Burstein (2019).[4] The framework is a real model of a closed economy, with the exception of international fiscal transfers, that allows for accumulation of physical capital, productivity-enhancing public capital, fiscal transfers, and innovative investment (Aghion and Howitt, 1992; Romer, 1990). This means that, in our framework, aggregate productivity and economic growth are endogenous to firms' decisions and therefore to economic policies such as public capital or fiscal transfers.

[4] We follow the exposition of Atkeson and Burstein (2019) where there is an overlay in the description of the model. A detailed description of model equations can be found in Appendix A.

2.1 Households

The economy is populated by an infinitely lived representative household with time-separable preferences over per capita consumption C_t/H_t and hours worked:

$$\max_{C_t, K_{t+1}, B_{t+1}, L_t^p, L_t^r} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t H_t \left(\log\left(\frac{C_t}{H_t}\right) - \kappa \frac{\left(\frac{L_t^r + L_t^p}{H_t}\right)^{1+\varphi}}{1+\varphi} \right) \quad \text{s.t.} \tag{1}$$

$$\begin{aligned}(1+\tau_t^C)C_t + I_t + B_{t+1} = (1-\tau_t^L)(W_t^p L_t^p + W_t^r L_t^r) + (1-\tau_t^{Corp}) r_t^k K_t + (1+R_{t-1})B_t \\ + D_t - T_t - \Xi^L(L_t^p, L_t^r, L_{t-1}^p, L_{t-1}^r),\end{aligned} \tag{2}$$

$$K_{t+1} = (1-\delta_K)K_t + I_t + T_t^K - \Xi^I(I_t, K_t), \tag{3}$$

where $\beta \in (0,1)$ and H_t denotes population, which grows at an exogenous rate g_H. The household can save in physical capital K_t, which rents to firms at rental rate r_t^k, and in government bonds B_t, with risk-free return R_t. Additionally, the household derives labor income from supplying production working hours L_t^p, paid at wage rate W_t^p, and research working hours L_t^r with wage rate W_t^r. Here, τ_t^L denotes the labor income tax rate, τ_t^C the consumption tax, τ_t^{Corp} is a corporate tax rate levied on capital returns, and T_t denote lump-sum taxes levied by the government. We assume that labor is subject to adjustment costs $\Xi^L(L_t^p, L_t^r, L_{t-1}^p, L_{t-1}^r)$. Finally, economy-wide firms' profits D_t are rebated lump-sum to the household.

The capital accumulation equation is given by (3), where δ_K is the depreciation rate of private physical capital and $\Xi^I(I_t, K_t)$ denotes capital adjustment costs. Capital accumulates over time due to two main forces. First, through the investment expenditures that the household finances with her own market income, I_t. Second, we consider government transfers, T_t^K, that the household can only use to build physical capital.[5]

We note that although we assume that the household has to spend government transfers T_t^K on building private capital, this does not mean that transfers do not distort the investment made by the household with the non-transfer income, I_t. Namely, under the production function we consider in section 2.2, the extra capital built with government transfers may come, in equilibrium, with a fall in the capital return r_t^k. Such a fall in the return to household savings can potentially crowd out investment I_t.

2.2 Production Sector

The production sector of the economy consists of three sub-sectors. First, a final good producer that bundles intermediate goods. Second, intermediate goods producers engage in innovative investment and use physical capital and production labor as production inputs. As we specify in more detail below, intermediate goods producers finance innovative investment through their own resources and also through government transfers that have to be spent on innovative activities, similar to the case of capital transfers. Third, there is a research good producer that uses research labor as factor input.

2.2.1 Final Good Producer

A competitive final good producer combines a continuum of differentiated intermediate goods using a constant elasticity of substitution (CES) production function to produce a

[5] In the data, transfers to build private capital are given to firms rather than to households. Alternatively, one could introduce in the model a firm that takes the role of a capital good producer and let the government give the transfers to that firm. However, since the representative household is the ultimate owner of all firms in this economy, that formulation and the one we consider here are equivalent.

final good Y_t:

$$\max_{y_t(z)} \quad Y_t - \sum_z p_t(z) y_t(z) M_t(z) \quad \text{s.t.} \quad Y_t = \left[\sum_z M_t(z) y_t(z)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}, \tag{4}$$

with $\rho > 1$. Above, $y_t(z)$ denotes the output of an intermediate producer with productivity index z, which has a price $p_t(z)$. $M_t(z)$ denotes the mass of intermediate goods with productivity index z at time t.

2.2.2 Intermediate Good Producers

There is a continuum of intermediate good producers that produce differentiated goods y using production labor l^p and physical capital k. We summarize the technology that is used in the production of an intermediate good at time t by its productivity index z. Moreover, we assume that public capital K_t^G improves the use of private resources by firms with efficiency χ, such that the production function for an intermediate good with productivity z is given by:

$$y_t(z) = z \left(K_t^G \right)^{\chi} k_t(z)^{\alpha} l_t^p(z)^{1-\alpha}, \tag{5}$$

with $\alpha \in (0,1)$. As in Atkeson and Burstein (2019), we assume that z has a countable support with grid elements z_n, and refer to the highest element in the grid for each intermediate good as the frontier technology for that good. Since capital and labor are flexible at the firm-level, the optimal allocation of production inputs maximizes per-period firms' variable profits associated with that good with productivity z, defined as:

$$\pi_t(z) \equiv p_t(z) y_t(z) - W_t^p l_t^p(z) - r_t^k k_t(z). \tag{6}$$

Intermediate-good producers can engage in innovative investment, which requires the purchase of a research good. Innovative investment results in the creation of a new product in the economy, as in Romer (1990), or in efficiency improvements of already existing products. As we show later, this leads to changes in aggregate productivity.

We allow both incumbent firms, those that produce at time t and were also producing at time $t-1$, and new entrants (that did not produce at time $t-1$) to invest in innovation. Innovative investment by entrants can only lead to the creation of products that are new to society.[6] Incumbent firms, additionally, can also improve the efficiency of the products that they already own. Furthermore, we assume that an exogenous fraction δ_0 of goods produced by incumbent firms exits the market each period. We describe the innovation process of incumbents and entering firms next.

Innovative investment by entering firms. We denote by M^e_{t+1} the measure of entrants that invest in innovation at time t. Each of these firms obtains at $t+1$ a frontier technology to produce a new intermediate good with productivity index z'. As in Atkeson and Burstein (2019), and similar to Luttmer (2007), we assume that z' is drawn from a distribution such that $\mathbb{E}_t z'^{\rho-1} = \eta_e Z_t^{\rho-1}/M_t$, where Z_t denotes aggregate productivity and M_t is the total measure of products available. Therefore, as in Atkeson and Burstein (2019), $Z_t^{\rho-1}/M_t$ is a measure of average productivity across existing intermediate goods. More precisely, these are defined as:

$$Z_t = \left(\sum_z z^{\rho-1} M_t(z)\right)^{\frac{1}{\rho-1}} \tag{7}$$

$$M_t = \sum_z M_t(z). \tag{8}$$

Finally, following Atkeson and Burstein (2019) and Acemoglu (2009), we assume that

[6]The model could also be easily adapted to include business stealing as is standard in quality ladder models (Klette and Kortum, 2004). See, for example, Atkeson and Burstein (2019).

an entering firm has to purchase $1/M_t$ units of the research good at time t to create a new firm with one product at time $t+1$. We denote by x_t^e the total amount of research goods purchased by M_{t+1}^e entering firms at time t. Note that this implies that $x_t^e = M_{t+1}^e/M_t$.

Denoting by $V_t(z)$ the value of an intermediate-good product with productivity index z at time t, the free-entry condition is given by:

$$\frac{1}{M_t} P_t^r = \mathbb{E}_t \frac{1}{1+R_t} V_{t+1}(z'), \tag{9}$$

where P_t^r denotes the price of the research good at time t. The left-hand side in (9) is the marginal cost of investing one additional unit in innovation $x_t^e(z)$, since this has a cost $P_t^r M_t^{-1}$ for an entering firm. The right-hand side is the expected discounted value of obtaining a product with productivity z' at time $t+1$ as a result of that innovative investment decision.

Innovative investment by incumbent firms. Incumbent firms can purchase research goods to create products that are new to society, as entering firms, or to improve the efficiency of the products that they already own.

First, consider the former case. An incumbent firm has the opportunity to invest $x_t^m(z)$ units of the research good at time t to create a new product at time $t+1$ with probability $h\left(x_t^m(z)/s_t(z)\right)$, where $s_t(z)$ is given by:

$$s_t(z) \equiv \left(\frac{z}{Z_t}\right)^{\rho-1}. \tag{10}$$

The variable $s_t(z)$ measures the productivity of a given product z relative to aggregate productivity Z_t at time t. It can be understood as a measure of size too, as in equilibrium it equals the share of revenue and inputs allocated to that product with productivity z.

Similar to the case of entering firms, we assume that the productivity index of the new product created by incumbent firms is drawn from a distribution such that $\mathbb{E}z'^{\rho-1} = \eta_m z^{\rho-1}$. Similarly, the aggregate quantity of research goods purchased by incumbent firms to create new products is given by $x_t^m = \sum_z M_t(z) x_t^m(z)$.

Second, consider the case of an incumbent firm that wishes to improve the productivity of a product with productivity index z that it already produces. Such firm purchases $x_t^c(z)$ units of the research good at time t and draws a new productivity index z' for its existing product – conditional on not exiting the market – from a distribution such that $\mathbb{E}z'^{\rho-1} = \zeta\left(x_t^c(z)/s_t(z)\right) z^{\rho-1}$. In a similar fashion to the previous cases, we define the aggregate quantity of this class of innovative investment $x_t^c = \sum_z M_t(z) x_t^c(z)$.

Under the previous assumptions, we can write the intertemporal problem associated with a product with productivity index z as:

$$V_t(z) = \max_{x_t^c(z), x_t^m(z)} (1-\tau_t^{\text{Corp}})\pi_t(z) - P_t^r\left(x_t^m(z) + x_t^c(z)\right) + T_t^{R,c}(z) + T_t^{R,m}(z) + \mathbb{E}_t \frac{1}{1+R_t} V_{t+1}(z'), \tag{11}$$

where $\pi_t(z)$ are per period variable profits as defined in (6). $T_t^{R,m}(z)$ and $T_t^{R,c}(z)$ are fiscal transfers for innovative investment provided by the government. We assume that aggregate transfers of this type, $T_t^{R,m}$ and $T_t^{R,c}$, are equally distributed across intermediate goods, such that $T_t^{R,m}(z) = T_t^{R,m}/M_t$ and $T_t^{R,c}(z) = T_t^{R,c}/M_t$. We discuss these next.

Fiscal transfers to innovation. The fiscal authority provides intermediate good producers with transfers that have to be spent on innovative investment, similar to the case of fiscal transfer to be used for capital accumulation. Namely, we consider that the total amount of the research good purchased for a type of innovative investment, $\{x_t^e(z), x_t^m(z), x_t^c(z)\}$, consists of the amount of goods that firms purchase using its own resources $\tilde{x}_t^j(z)$ plus the amount of goods purchased using fiscal transfers $T_t^{R,j}(z)$:

$$P_t^r x_t^j(z) = P_t^r \tilde{x}_t^j(z) + T_t^{R,j}(z), \quad \text{with} \quad j \in \{e, m, c\} \tag{12}$$

In line with the case of fiscal transfers, we also note here that fiscal transfers $T_t^{R,j}(z)$ can distort firms' innovative investment decisions in equilibrium. This follows since a greater demand for research goods arising from an increase in transfers may lead to an increase in the price of the research good, leading to a fall in the amount that firms spend on innovative investment using their own resources.

2.2.3 Research Good Producer

A representative research good producer hires research labor $L_t^{r,d}$ to produce the research good used in the innovation process according to:

$$Y_t^r = A_t^r \left(K_t^G\right)^\chi Z_t^{\phi-1} L_t^{r,d}. \tag{13}$$

Above, A_t^r can be interpreted as a stock of freely available scientific progress, which we assume to grow at an exogenous rate g_{A_r}. Moreover, as in the case of the production of intermediate goods, we assume that public capital K_t^G improves the efficiency with which research labor is used.[7] Finally, the term $Z_t^{\phi-1}$, with $\phi \leq 1$, follows Jones (2002). It represents intertemporal knowledge spillovers. Namely, since $\phi \leq 1$, increases in aggregate productivity Z_t reduce the efficiency of research labor, capturing the notion outlined in Bloom et al. (2020) that "ideas are getting harder to find".

The research good is sold at price P_t^r to intermediate good produces engaging in innovative investment, such that the research good producer solves the following intratemporal problem:

$$\max_{L_t^r} \quad P_t^r Y_t^r - W_t^r L_t^{r,d} \quad \text{s.t.} \quad Y_t^r = A_t^r \left(K_t^G\right)^\chi Z_t^{\phi-1} L_t^{r,d} \tag{14}$$

[7] There might be uncertainty about the extent to which the elasticity of research output to public capital must be the same as for general production. Hence, we have also considered simulations, available upon request, assuming that public capital does not directly increase the efficiency of research labor. We have found that this assumption has negligible effects on our baseline results.

2.3 Government

The government consists of a fiscal authority. The government raises revenue from labor income taxes, consumption taxes, corporate taxes, and lump-sum taxes. In addition, we allow for the possibility that the government receives international transfers T_t^{EU}. It uses the revenue and government debt B_{t+1}^s to finance interest payments on public debt and expenditures. Government expenditures consist of public investment I_t^G and transfers to firms dedicated to either purchasing physical capital T_t^K or research goods $T_t^{R,j}$. Therefore, the budget constraint of the government is given by:

$$
\begin{aligned}
I_t^G + B_t^s(1+R_{t-1}) + T_t^{R,e} + T_t^{R,m} + T_t^{R,c} + T_t^K = B_{t+1}^s& \\
+ \tau_t^L(W_t^r L_t^r + W_t^p L_t^p) + \tau_t^C C_t + \tau_t^{\mathrm{Corp}}&\Big(\sum_z M_t(z)\pi_t(z) + r^K K_t\Big) + T_t^{\mathrm{EU}} + T_t, \\
\text{with}\quad I_t^G = K_{t+1}^G - (1-\delta^G)K_t^G&,
\end{aligned}
\tag{15}
$$

where δ^G denotes the depreciation rate of public capital.

2.4 Equilibrium, Market Clearing, and Productivity Dynamics

A *competitive equilibrium* is a sequence of consumption and hours $\{C_t, L_t^r, L_t^p\}_t$, private and innovative investment $\{I_t, x_t^e, x_t^m, x_t^c\}_t$, such that given prices $\{W_t^r, W_t^p, P_t^r, r_t^k, R_t, q_t\}_t$ and a sequence for public capital $\{K_t^G\}_t$, transfers $\{T_t^{\mathrm{EU}}, T_t^K, T_t^{R,e}(z), T_t^{R,m}(z), T_t^{R,c}(z)\}_t$, and taxes $\{\tau_t^L, \tau_t^C, \tau_t^L, \tau_t^{Corp}, T_t\}_t$ such that households and firms optimize and markets clear:

1. The labor market clears if $L_t^p = \sum_z M_t(z) l_t^p(z)$ and the amount of research hours supplied by the household equals the research hours demanded by the research good producer, $L_t^r = L_t^{r,d}$.

2. The capital market clears if $K_t = \sum_z M_t(z) k_t(z)$

3. The market for research goods clears if the innovative investment by entering firms and incumbent firms equals the production of the research good:

$$Y_t^r = x_t^e + x_t^m + x_t^c \tag{16}$$

4. The aggregate resource constraint of the economy is given by:

$$C_t + I_t + I_t^G + \Xi^L(L_t^p, L_t^r, L_{t-1}^p, L_{t-1}^r) + \Xi^I(I_t, K_t) = Y_t + T_t^{EU} \tag{17}$$

5. The bond market clears if government's bond supply equals households' bond demand: $B_t^s = B_t$.

6. The market for intermediate good producers clears by Walras's law.

Similarly, we define a *Balanced Growth Path* (BGP) as a competitive equilibrium where all variables grow at constant rates. In Appendix A.4 we provide a detailed description of detrended variables and associated equilibrium conditions in terms of stationary variables.

2.5 Aggregation and Productivity Dynamics

In Appendix A.3 we show that in equilibrium, under the assumptions made on the innovation process of firms together with constant markups, aggregate output can be written as:

$$Y_t = Z_t (K_t^G)^{\chi} (K_t)^{\alpha} \left(L_t^p\right)^{1-\alpha}, \tag{18}$$

where aggregate productivity Z_t is defined in (7).

The expression for aggregate output (18), together with expression for Z_t, makes clear that firm-level innovative investment – and policies that affect it – lead to endogenous changes into aggregate productivity Z_t, and hence on output Y_t.

We can see this point more clearly by deriving an expression for the dynamics of aggregate productivity following Atkeson and Burstein (2019). First, we consider the dynamics for the total measure of products available, M_t, which is given by:

$$M_{t+1} = (1 - \delta_0)M_t + x_t^e M_t + h\left(x_t^m\right) M_t. \tag{19}$$

The above expression states that the total measure of products available in $t+1$, M_{t+1}, is governed by three forces. The first one – the first term on the right-hand side of (19) – corresponds to the exogenous exit of products from the market. That is, only a fraction $1 - \delta_0$ of existing products in t survive to the next period. The second force – second term in equation (19) – corresponds to the mass of entering firms, $M_{t+1}^e = x_t^e M_t$, which engage in innovative investment at time t to create a new product at time $t+1$. Similarly, the last term corresponds to the fraction of incumbent firms that engage in innovative investment to create new products.

Following a similar logic, we can derive the dynamics of aggregate productivity Z_t, given by:

$$Z_{t+1}^{\rho-1} = (1 - \delta_0)\zeta(x_t^c)M_t \frac{Z_t^{\rho-1}}{M_t} + \eta_m h(x_t^m) M_t \frac{Z_t^{\rho-1}}{M_t} + \eta_e x_t^e M_t \frac{Z_t^{\rho-1}}{M_t} \tag{20}$$

The level of productivity next period, Z_{t+1}, depends on three terms determined by the innovative investment of incumbents and entrants. The first term on the right-hand side of equation (20) is the average productivity $t+1$ of products that were already produced at time t by incumbent firms that did not exit the market. The second term corresponds to the average productivity of new products at time $t+1$ that results from innovative investment incurred by incumbent firms at time t. The final term in equation (20) is the average productivity of new products in the economy resulting from innovative investment of entering firms.

Taking logs in equation (20) and rearranging we can then express aggregate productivity growth, $g_{Z,t} \equiv \log Z_{t+1} - \log Z_t$, as:

$$g_{Z,t} = \frac{1}{\rho - 1} \log\left((1 - \delta_0)\zeta(x_t^c) + \eta_m h(x_t^m) + \eta_e x_t^e\right), \tag{21}$$

which makes explicit the dependence of productivity growth on innovative investment decisions of incumbent and entering firms.

3 Calibration

We calibrate the model to the Spanish economy. The calibration period is 2000-2019, starting shortly after the creation of the Euro Area and ending right before the COVID-19 crisis. One period in the model corresponds to one year. We mainly draw from two data sources to calibrate the model. First, we obtain aggregate data from the National Statistical Office of Spain (*Instituto Nacional de Estadística*, *INE*). Second, we rely on the Central Balance Sheet Data Office (*Central de Balances*), maintained by *Banco de España*, to obtain the firm-level data used in the calibration of the innovation process of firms.

3.1 Household Sector

We set the inverse of the Frisch elasticity φ to be equal to one, in the range of the estimates provided in Chetty et al. (2011). The parameter governing the disutility from labor, κ, is set to match an average employment rate within the working-age population in Spain of 64% over the period. We set the time-discount factor, β, to target an annualized interest rate of 4.5% at the steady state. The population growth, g_H, is set to 0.6%, in line with the average population growth in Spain over the sample period.

As regards the parameters affecting the capital accumulation process, we first assume a functional form for the capital adjustment costs $\Xi^I(I_t, I_{t-1}, K_t)$ similar to Christiano et al. (2011):

$$\Xi^I(I_t, K_t) = \frac{\sigma_I}{2}\left(\frac{I_t}{K_t} - (\delta_K + \exp(g_Y) - 1)\right)^2 K_t, \tag{22}$$

where g_Y denotes the constant growth rate of output at the BGP. We set σ_I equal to 17 to target the elasticity of Tobin's Q to investment documented in Eberly et al. (2008).[8]

Next, we set the depreciation rate of private physical capital, δ_K, to target a private-investment-to-GDP ratio of 21% in steady state. This results in $\delta_K = 3.3\%$.

We assume that labor adjustment costs have a similar functional form:

$$\Xi^L(L_t^p, L_t^r, L_{t-1}^p, L_{t-1}^r) = \frac{\sigma_L}{2}\left(\log(\frac{L_t^r/H_t}{L_{t-1}^r/H_{t-1}})^2 + \log(\frac{L_t^p/H_t}{L_{t-1}^p/H_{t-1}})^2\right). \tag{23}$$

We calibrate the parameter governing the labor adjustment costs to match the evidence on public investment multipliers for Spain presented in de Castro and Hernández de Cos (2008). Since an important component of NGEU funds is public investment, we regard this as a reasonable starting point to provide a more general assessment of the impact of the NGEU program. More precisely, we consider an AR(1) shock to public investment to closely match the empirical impulse response of public investment presented in that paper. We then choose the labor adjustment cost, σ_L, to target a 5-year (the longest horizon provided in de Castro and Hernández de Cos (2008)) cumulative public investment multiplier of 0.69. This results in $\sigma_L = 0.10$.

[8]See, for example, Christiano et al. (2011) and de Ferra et al. (2020) for papers following a similar approach. In addition, in Appendix B, we consider a simulation without adjustment costs to capital. The main results are barely affected.

3.2 Government

We assume that the stock of public capital depreciates at the same rate as the private capital, meaning that $\delta_G = 3.3\%$, as in Stähler and Thomas (2012).[9] Given a value for the depreciation rate, we then choose the stock of public capital to target a public-investment-to-GDP ratio of 3.7% in steady state.

The range of estimates of the efficiency of public capital, χ, is rather wide (cf. Bom and Ligthart, 2014). Therefore, in our simulations, we will consider two possible values for this parameter, $\chi \in \{0.05, 0.15\}$, which are towards the lower end and upper end of available estimates in Bom and Ligthart, 2014, respectively.

Regarding tax rates, we proceed as follows. We first normalize the consumption tax rate to 0.[10] We then set the corporate tax rate to target a ratio of firms' tax payments to profits. More precisely, we obtain from the quarterly national accounts the ratio of direct taxes paid by non-financial corporations as a share of their gross operating surplus. In the data, this ratio equals to 9.17%. This procedure results in a corporate tax rate of $\tau^{\text{Corp}} = 0.29$ in the model, very close to the actual general corporate tax rate of 25% in Spain.

Next, we set international transfers, transfers to firms, and public debt equal to zero at the BGP. We set the labor income tax τ^L equal to 13%, to match the effective income tax in Spain in 2019, as reported by the Spanish Tax Agency.[11] Given these fiscal variables, we then set the lump-sum taxes, T_t, to ensure that the government budget constraint holds at the BGP.

[9] As a robustness check, in Appendix B we consider an alternative simulation under the assumption that the depreciation rate of public capital is lower than the depreciation rate of private capital, along the lines of Ramey (2020). This has little impact in our results.

[10] All tax rates are constant in the BGP. Therefore, the only first-order condition where the consumption tax is in the intratemporal condition characterizing labor supply. It does so in a symmetric way to the labor income tax, such that increasing one is equivalent to decreasing the other.

[11] See the Tax Revenue Annual Report of the 2019 Fiscal Year (`https://sede.agenciatributaria.gob.es/Sede/en_gb/estadisticas/recaudacion-tributaria/informe-anual/ejercicio-2019.html`) prepared by the Spanish Tax Agency (*Agencia Tributaria*).

Parameter	Description	Value	Target / Source
Households			
κ	Disutility from labor	1.61	$L/H = 0.64$
φ	Frisch elasticity	1	Chetty et al. (2011)
β	Discount factor	0.97	$R = 4.5\%$
g_H	Growth rate Pop.	0.6%	INE
δ_K	Depreciation capital	3.3%	$I/Y = 0.20$
σ_I	Capital adj. cost	17	Eberly et al. (2008)
σ_L	Labor adj. cost	0.10	de Castro and Hernández de Cos (2008)
Government			
τ^l	Labor income tax rate	0.13	Avg. lab. inc. tax in Spain
τ^{Corp}	Corporate profit tax rate	0.29	Firms' taxes / profits = 9.17%
δ_G	Depreciation public capital	3.3%	Same as δ_K
χ	Efficiency public capital	$\{0.05, 0.15\}$	Bom and Ligthart (2014)
Production			
ρ	Elasticity Substitution	4	Broda and Weinstein (2006)
ϕ	Intertemp. Externatility	-1.6	Fernald and Jones (2014)
g_{Ar}	Exogenous Prod. Growth	0.07%	$g_Y = 1.6\%$
α	Capital exponent prod. fct.	0.43	Labor share $= 0.55$
μ	Markup	1.05	$K/Y = 4.23$
Innovation			
δ_0	Exit rate	0.05	
η_e	Prod. step entrant	1.6	
η_m	Prod. step incumb.	0.74	See text for a discussion
$\{h_0, h_1\}$	Fct. innov. new prod. incumb.	$\{0.4, 0.5\}$	
$\{\zeta_0, \zeta_1, \zeta_2\}$	Fct. innov. exist. prod. incumb.	$\{0.9, 0.5, 0.5\}$	

Table 1: Calibration

Notes: List of calibrated parameters. See text for a discussion on targets, values, and data used.

3.3 Production and Innovation

We set the elasticity of substitution across intermediate goods equal to 4, in line with the estimates of Broda and Weinstein (2006). Next, we set the exogenous growth rate of A equal to 0.07% to target an annual growth rate of output of 1.6% at the BGP, the average growth rate GDP in our sample period. We set α to target a labor share of 0.55, and the markup μ in the model is set to target a capital-to-output ratio of 4.23, as in

Arencibia Pareja et al. (2018).

In the production function of research goods (13) ϕ governs the intertemporal externality of technological progress. There is ample evidence showing that this parameter should be negative (e.g. Bloom et al. (2020)), meaning that a current high level of productivity reduces the productivity of new innovation, capturing the idea that "ideas are getting harder to find". In light of this, we follow Fernald and Jones (2014) and Atkeson and Burstein (2019) and set ϕ equal to -1.6.[12]

Our calibration strategy for the innovation process of firms closely follows Atkeson and Burstein (2019) whenever possible. Namely, we next posit the following functional forms for the innovation functions of incumbents for new products, $h(x_t^m)$, and for existing products, $\zeta(x_t^c)$:

$$h(x_t^m) = h_0(x_t^m)^{h_1} \tag{24}$$

$$\zeta(x_t^c) = \zeta_0 + \zeta_1(x_t^c)^{\zeta_2}. \tag{25}$$

Therefore, there are eight remaining parameters to be calibrated. We need to calibrate the productivity steps of new products for entrants and incumbents, η_e and η_m; the exogenous exit rate of products from the economy, δ_0; and the parameters governing the innovation function for new products (h_0 and h_1) and for continuing products (ζ_0, ζ_1, and ζ_2).

Our calibration strategy for these parameters is as derived in Atkeson and Burstein (2019). Namely, let f_e be the fraction of products produced by entering firms (M_e/M), and let f_m and $f_c = 1 - f_e - f_m$ be defined analogously as the fraction of, respectively, new and continuing products produced by incumbent firms. Additionally, let s_e denote the

[12] Bloom et al. (2020) provide an ample range of estimates depending on different innovation categories and firms. Their aggregate estimate is approximately -2 (see the discussion in Jones (2022)), very close to our baseline value of ϕ.

aggregate size of products produced by entering firms (as defined by aggregating across entering firms in equation (10)), and define s_m and $s_c = 1 - s_e - s_c$ analogously for new and continuing products produced by incumbent firms. Finally, define the average size for each type of product as $\text{avsize}_j = {}^{s_j}/_{f_j}$ for $j \in \{e, m, c\}$. Let g_M denote the growth rate of the M.

The parameter values for $\{\delta_0, \eta_m, \eta_e\}$ and the steady state values of $\{h(x_m), \zeta(x_c), x_e\}$ are pinned down by the following set of equations:

$$h(x_m) = f_m exp(g_M) \tag{26}$$

$$x_e = f_e exp(g_M) \tag{27}$$

$$\zeta(x_c) = \text{avsize}_c \frac{exp\left((\rho - 1)g_Z\right)}{exp g_M} \tag{28}$$

$$\eta_m = \text{avsize}_m \frac{exp\left((\rho - 1)g_Z\right)}{exp g_M} \tag{29}$$

$$\eta_e = \text{avsize}_e \frac{exp\left((\rho - 1)g_Z\right)}{exp g_M} \tag{30}$$

$$\delta_0 = 1 - f_c exp(g_M). \tag{31}$$

The previous set of equations can be derived by using the law of motion for M (19), the law of motion for Z (20), the expressions for the total measure of new products produced by entrants and incumbents ($x_{e,t} = {}^{M_{e,t+1}}/_{M_t}$ and $h(x_m) = {}^{M^m_{t+1}}/_{M_t}$) and the assumed distributions for productivity draws of new products by entering ($\mathbb{E} z'^{\rho-1} = \eta_e {}^{Z_t^{\rho-1}}/_{M_t}$) and incumbent firms ($\mathbb{E} z'^{\rho-1} = \eta_m z^{\rho-1}$).

Given these, we can then use the firms' first-order conditions to obtain $h'(x_m)$ and $\zeta'(x_c)$, as derived in Appendix A:

$$\eta_e = \eta_m h'(x_m) \tag{32}$$

$$\eta_e = (1 - \delta_0)\zeta'(x_c) \tag{33}$$

Note that we already have a value for x_e from equation (27). Given this, we then pin down $x_c + x_m$ by targeting the aggregate R&D expenditures to GDP ratio of 1.41%, obtained from *INE*. This target pins down x_c and x_m jointly, but not separately. As to separate x_c and x_m we use the same logic as in Atkeson and Burstein (2019). Namely ${}^{P^r x_c}/_{Y}$ must lie between an upper bound ${}^{P^r(x_c+x_m)}/_{Y}$ and lower bound found by obtaining the value of a product to a firm when innovation on continuing products is zero. [13] Following this strategy, x_c is found by setting ${}^{P^r x_c}/_{Y}$ to the midpoint between these two bounds. Given x_c and x_e, this also implies a value for x_m.

Given x_m, together with the values for $h(x_m)$ and $h'(x_m)$, we can then pin down the values for h_0 and h_1 by making use of the functional form (24). We next set ζ_2 equal to 0.5 which is the midpoint of admissible values for this parameter according to Atkeson and Burstein (2019). Given ζ_2 and x_c we can then pin down ζ_0 and ζ_1 using the functional form (25), given that we have already obtained the values for $\zeta(x_c)$ and $\zeta'(x_c)$.

In order to implement this calibration strategy we use two sources of firm-level data, in addition to the aggregate R&D expenditures data mentioned above. First, we use data from the Central Business Directory of *INE* (*DIRCE*) to obtain a growth rate of the number of firms, g_M, of roughly 1%. Second, we rely on firm-level data from the Central Balance Sheet Data Office. Atkeson and Burstein (2019) use data on employment and the number of establishments. Since our data set only contains information at the firm level but not at the establishment level, we use data on production rather than on the number of establishments. Yet, we obtain data moments – which we describe next – that are reasonably close to the moments computed by these authors, as presented in Table 3 of the online appendix of that paper.

Using these data we measure $F_e = 0.02$ as the fraction of production today that is accounted by the production of new firms, $F_m = 0.05$ as the fraction of production today

[13] A derivation of these bounds can be found in Appendix C.3. of Atkeson and Burstein (2019).

that is accounted by the increase in production of incumbent firms, and $F_c = 0.93$ by imposing $F_c = 1 - F_e - F_m$. We then measure $S_e = 0.03$ as the share of employment that corresponds to new firms, $S_m = 0.03$ as the share of employment that corresponds to the increase in employment of incumbent firms, and $S_c = 0.94$ by imposing $S_c = 1 - S_e - S_m$.

Before moving to aggregate results, we follow Schmitz (2021) and compare our model outcomes with R&D firm-level data from *Encuesta sobre Innovación en las empresas*, a survey conducted by *INE*, as a way to validate the predictions implied by our model. The observed aggregate innovation intensity – measured as the innovation expenditures over sales – in the data is roughly 2%. In our model, this same statistic is equal to approximately 2.2%.

This increases our confidence in the suitability of the model to evaluate the macroeconomic consequences of the NGEU program, which is designed to have an important innovation component. In our aggregate simulations, to which we turn next, we will show that the model not only does a reasonable job in matching firm-level data moments but also delivers an elasticity of productivity to innovation expenditures in line with the available empirical evidence.

4 Simulation Exercises

We next use the model to assess the aggregate impact of the NGEU program on the Spanish economy, considering alternative allocations of funds, and compute the implied fiscal multipliers associated with the fiscal program. We compute the transition paths that follow the implementation of the NGEU program within our model under the assumption of perfect foresight. We first describe the mapping between the data on NGEU funds and the model and our assumptions regarding the funding of the program, and later present our main quantitative results and counterfactuals.

4.1 Mapping the NGEU funds to the model

We focus on the key instrument of the NGEU program, the Recovery and Resilience Facility (RRF), approved by the European Union in 2020 to foster a persistent recovery of EU member states. In this context, Spain has been allocated funds equivalent to roughly 6.4% of 2019 GDP in the form of grants. In order to channel these funds into the economy, the Spanish Government has enacted the Recovery, Transformation, and Resilience Plan (*Plan de Recuperación, Transformación, y Resilencia*, or *PRTR*). [14] In our framework, the reception of grants from the NGEU program takes the form of international transfers T_t^{EU}, incorporated into the budget constraint of the government (15).

We map the different projects included in the *PRTR* to the instruments available in the model following the breakdown provided in Fernández-Cerezo et al. (2023). Of the overall size of the program, 13% of the funds are allocated to public consumption expenditures; we abstract from this share of the funds since our focus is on policies that directly affect output over the medium term through the supply side of the economy. The remaining 87% of the funds is allocated between public investment to build and improve infrastructures such as ports, and road and rail networks (47%) – in our model, I_t^G – and transfers to private firms (40%), which in turn are split into roughly equal amounts between transfers targeted to build private physical capital (T_t^K) and transfers targeted to improve the technology with which firms operate (T_t^R). Regarding the latter, we assume that transfers for innovation investment are equally split between the three different types ($T_t^{R,e} = T_t^{R,m} = T_t^{R,c}$).

Within the NGEU program, examples of innovation investment are, to name a few, the programs designated to promote the development of technologies that facilitate energy

[14] The original allocation of grants to Spain amounted to €69.5 bn (5.9% of 2019 GDP), which was later increased by an additional €7.7 bn after revising the magnitude of the pandemic recession in Spain and by roughly €2.6 bn as a consequence of the Ukraine war. Furthermore, a similar amount of funds is available for Spain to be requested in the form of loans. We abstract from these potential funds in this paper.

storage, the "Space Technology Program" (*PERTE Aerospacial*) which aims to foster R&D development in the Spanish aerospace industry, or the "Digital Kit" (*Kit Digital*) program targeted to small firms in order to help them approach the digital frontier.[15] Next to these, an example of capital transfers is the funds designated to build electric car battery factories within the "Strategic Project for the development of electric vehicles" (*PERTE VEC*). Importantly, most of these funds are allocated as grants to firms, in line with our modeling of fiscal transfers.

Finally, regarding the shape and time of the spending paths, we assume for simplicity that funds are received and spent uniformly over an eight-year period, starting in 2021 and ending in 2028.[16] [17]

We plot in Figure 1 the implied path for transfers and expenditures in the model, as a percentage of 2019 Spanish GDP. International transfers (panel a) amount to nearly 0.7% of GDP per year. The bulk of these transfers is used to increase public investment (panel b). Note that transfers expire in 2028, but public investment remains permanently higher than in 2019; this happens because we assume that once the new public capital has been built between 2021 and 2028, when the NGEU program is finished, the government raises investment to cover the depreciation costs of the newly built capital.[18] The remaining funds are equally split between transfers to build physical capital (panel c) and transfers to increase innovative investment (panel d).

In our simulations, the NGEU grants are repaid partially and indirectly. There is no

[15]For examples of programs for the development of energy storage see https://www.idae.es/ayudas-y-financiacion/primera-convocatoria-para-proyectos-de-id-de-almacenamiento-energetico-dentro

[16]The NGEU funds are initially designed to finish in 2026. However, some programs under the umbrella of the *PRTR* have already been extended to 2028. See, for example, the case of the funds designated to the design and production of electric vehicles https://www.lainformacion.com/clima/gobierno-extiende-2028-plazo-perte-vec/2881737/

[17]In Appendix B we consider the effects of the NGEU program under alternative spending profiles. There, we show that the main results of our baseline specification remain robust to considering increasing or decreasing spending paths, with only marginal effects on the shape of the transition.

[18]We deem this assumption as natural. It says, for example, that if the government builds new highways it does not let them rot until they become obsolete once the NGEU program ends.

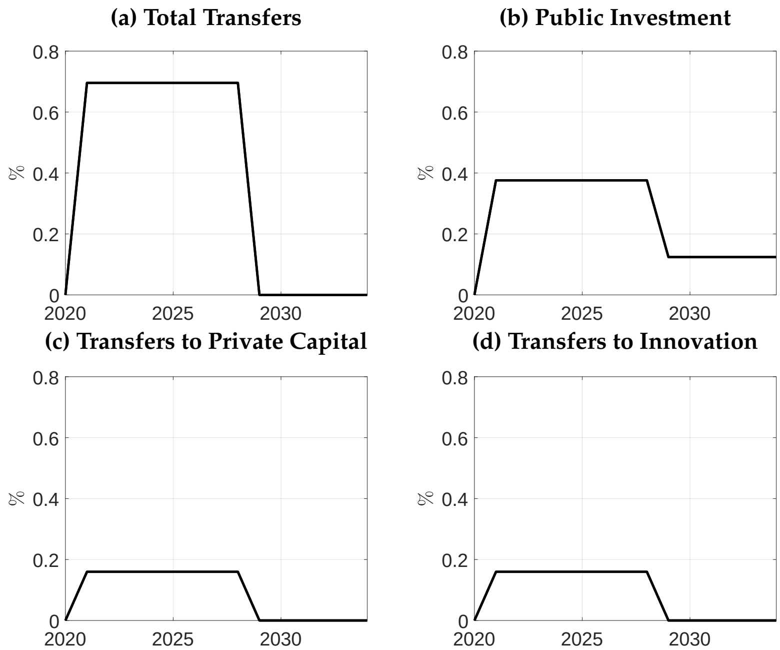

Figure 1: Fiscal Mix

Notes: This figure shows the path for policy variables in our simulation of the NGEU program.

explicit repayment, but EU member states must contribute to the union-wide budget according to their share of GDP, which is then used by the EU to pay back the debt issued to finance the NGEU funds between 2028 and 2058.[19] The share of union-wide NGEU grants received by Spain is roughly 20%, while the Spanish economy represents approximately 10% the EU GDP. Therefore, in our simulations, we assume that between 2028 and 2058 half of the international transfers received are paid back by raising both labor income taxes and consumption taxes.[20] We also assume that the government raises these same tax rates to finance the additional public investment required to cover the depreciation costs of the newly built public capital (see panel b, Figure 1).

[19] See https://commission.europa.eu/strategy-and-policy/eu-budget/eu-borrower-investor-relations/nextgenerationeu_en

[20] See Pfeiffer et al. (2022) for a similar assumption. Contrary to that paper, however, we assume that the additional revenue required to pay back grants is raised through distortionary taxation, rather than through lump-sum taxes.

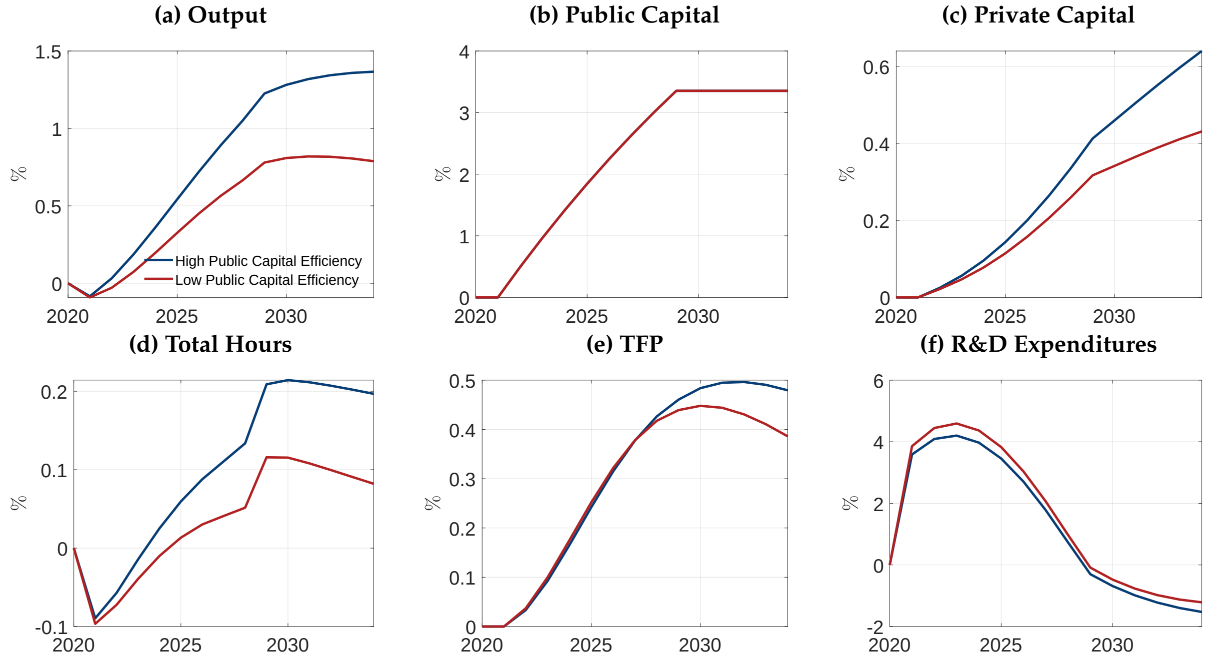

Figure 2: Aggregate Consequences of the NGEU funds

Notes: Impulse response functions to the path of the NGEU program, described in detail in section 4.1. Blue lines plot the path under a high efficiency of public capital, $\chi = 0.15$, and red lines under a lower efficiency of public capital, $\chi = 0.05$.

4.2 Aggregate Consequences of the NGEU funds

Figure 2 shows the responses of aggregate variables to the NGEU funds in our model, implemented as described in the previous section. Since public investment accounts for a important share of the program, we provide responses with the two different values of public capital efficiency in terms of increasing private productivity. The blue lines show the case with relatively high efficiency of public capital, $\chi = 0.15$. This would be our baseline simulation. The red lines display the responses with lower efficiency of public capital, $\chi = 0.05$.

Panel (a) shows that the NGEU funds have a quantitatively relevant effect on the level of output – as defined in (18) – and on output growth. In panel (a) we can observe that by 2028, when transfers expire in our simulation, output is between 0.7% and 1.1% higher relative to the baseline without NGEU funds. The average growth rate of output over that period increases by roughly $0.08 - 0.13$ percentage points. The increase in output does not stop in 2028, even though fiscal variables remain constant thereafter, but it continues to be higher for several more years until output stabilizes at a new level.[21]

Panels from (b) to (f) display the evolution of the different production inputs. Public capital (panel b) increases slowly over time as a result of higher public investment, stabilizing after 2028 at a new level that is permanently higher. Private capital (panel c) builds up much more slowly and takes a longer to stabilize to new levels, as households generate savings at a slower pace to ensure a smooth consumption path. We observe here that the assumed efficiency of public capital has a non-negligible effect on the path of private capital. Namely, a more productive public capital allows firms to use their private resources more efficiently, increasing firms' demand for capital and resulting in a higher accumulation of capital.[22]

Panel (d) shows the response of total hours worked. There is a small decline in total hours worked on impact due to a positive wealth effect from the NGEU program, which induces households to choose more leisure. Specifically, the transfers received from the EU make households feel wealthier, thereby reducing their labor supply. Over time, productivity increases (panel e), prompting households to work more as a result of a substitution effect. As productivity continues to increase, hours gradually rise over time. Finally, once productivity stabilizes after the NGEU program ends, the substitution effect weakens, leading to a reduction in hours worked.[23]

Aggregate productivity Z_t is displayed in panel (e). In both cases, it increases steadily until 2028 and by a similar amount to the response of private capital. Furthermore, we

[21] There are permanent effects on the level of output since we consider that the government does not let the newly built public capital depreciate, recall Figure 1.

[22] In Appendix B we further discuss the crowding out of private investment in physical capital and innovation, induced by the increase in prices that the fiscal stimulus generates.

[23] We are thankful to an anonymous referee for suggesting this interpretation.

observe that in this case the blue and red lines closely track each other until the end of our simulated NGEU program, suggesting that fiscal transfers destined for innovative uses are one of the main drivers of the change in productivity, with public capital having a more limited role on this variable.

The behavior of productivity Z_t is a key component explaining the output gains observed in panel (a) since productivity gains map one to one into output (recall the expression for aggregate output (18)). Namely, for example in the case of $\chi = 0.15$ displayed with blue lines, productivity increases by roughly 0.49% in 2031. In that same year, output increases by roughly 1.32%. This means that, without the productivity gains, output would have increased by only 0.83%, reducing the impact of the NGEU stimulus to a 62% of our baseline simulation. The fact that we account for the endogenous response, shown to be a key driver of the aggregate impact, also sets apart the current paper from previous analysis of the NGEU program (see, for example, Pfeiffer et al. (2022) and Fernández-Cerezo et al. (2023)).

The response of aggregate productivity predicted by the model is not only quantitatively relevant but also in line with the available empirical evidence. To see this, it is useful to first consider the response of R&D expenditures (panel f), which drives the response of productivity (recall the equation determining the law of motion of Z, (20)). R&D expenditures peak at roughly 4.20%, which later leads to a peak response of 0.49% in the level of Z.[24] This gives us a peak-to-peak elasticity of productivity to R&D expenditures of approximately 0.11. This can be compared, for example, with Hall (2011), which documents an elasticity of productivity to innovation that primarily ranges between 0.09 and

[24]We note that R&D expenditures also fall after 2028. Looking at the production of function or research goods, $Y_t^r = A_t^r \left(K_t^G\right)^\chi Z_t^{\phi-1} L_t^{r,d}$, we see that such a fall could be due to either a drop in research labor ($L_t^{r,d}$) or to the negative externality captured in $Z_t^{\phi-1}$. We have verified that research labor is still positive during those years, and hence the reason for the fall is the lagged increase in the overall level of productivity (Z_t), which exerts downward pressure on the efficiency of current research output. Such effect would be in line with the empirical evidence presented in the Bloom et al. (2020), who argue that as productivity levels increase there is a slowdown in the production of innovation.

0.13 in different sectors of Western European countries. Additionally, Moran and Queralto (2018) documents – by means of a VAR – a peak-to-peak elasticity of TFP to R&D expenditures of approximately 0.10 using US data, and for the remaining OECD countries this elasticity ranges from 0.05 up to 0.20 for the top R&D-performing countries.

To further emphasize the role played by innovative investment and productivity, we consider in Appendix B an alternative calibration of the parameters of the innovation functions where the elasticity of productivity to R&D is increased up to 0.15, towards the upper range of empirical estimates. Under this calibration, we obtain that the economic impact of the NGEU would be meaningfully larger, as now innovation expenditures translate to a larger extent into productivity gains. This result not only reinforces that the endogenous response of productivity to the fiscal stimulus is a key component driving the output gains of the fiscal stimulus, but also that an efficient transformation of innovation expenditures into productivity gains is central for the stimulus to be effective.

4.3 Alternative Allocation of NGEU funds

The previous section has shown that the current allocation of NGEU funds within the context of the Spanish economy can be expected to have significant positive effects on economic activity and that a key driver of this result was productivity. We next use the model as a laboratory to explore counterfactual simulations that provide some hindsight about which of the components of the NGEU is more effective in raising output and how the economy would fare under alternative allocations of the funds.

We consider three counterfactual allocations of the NGEU funds. Namely, instead of our baseline allocation of funds, we next assume that all transfers displayed in panel (a) of Figure 1 are allocated to one of the three policy instruments that we consider: either (i) public capital, (ii) transfers to fund an increase of private physical capital, or (iii) transfers

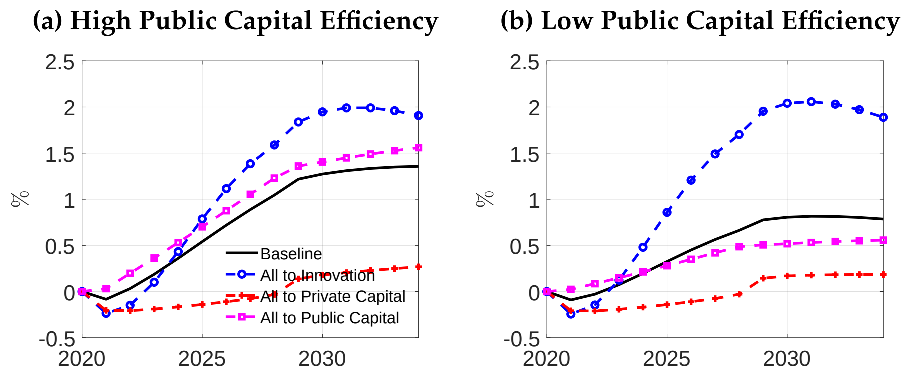

Figure 3: Alternative allocations of NGEU funds

Notes: Path for output under alternative allocations of the NGEU funds. The left panel shows the effects with high efficiency of public capital, $\chi = 0.15$, and the right panel for a lower efficiency of public capital, $\chi = 0.05$. The black lines display our baseline simulation, see section 4.2. The remaining lines consider counterfactual simulations, where all NGEU funds are allocated to either public capital (pink line, squares), innovation transfers (blue line, circles), or private capital transfers (red line, circles).

towards innovative investment.[25]

We show the response of output in each of these scenarios in Figure 3, where the left panel shows the responses of output for the case of a high efficiency of public capital, and the right panel displays the case of a lower efficiency of public capital. In both panels, we show with black lines our baseline allocation of funds. Pink lines with squares show the path for output under the alternative allocation where all funds are used for public capital. Blue lines with circles show the output response when all funds go to finance innovative investment. Finally, red lines with circles display the output path under the assumption that all transfers are used to build private physical capital.

Consider first the case where all transfers are allocated to the provision of public capital. In this scenario, we observe that the implied response of output is relatively close to the path under the baseline allocation of funds (the pink lines are similar to the black lines). This is true for both levels of public capital efficiency. Therefore, it seems that

[25] As to make sure that these counterfactuals are comparable to our baseline, in all these alternative scenarios we assume that once transfers end, the fiscal authority keeps the same path for public investment as plotted in panel (b) of Figure 1.

moving from the current allocation of NGEU funds to the alternative where all transfers are used for public investment would not significantly change the effects on aggregate output.

Next, focus on the case where all transfers are allocated to fund innovative investment (blue lines with circles). The output effects of transfers to innovative investment would be a bit smaller over the first years than in the baseline and in the case of the allocation of all funds to public capital. This is so because the gains through increases in productivity take some time to materialize (recall Figure 2, panel e). However, output gains become larger over time, surpassing those of the baseline and more notably when the efficiency of public capital is low. In other words, according to our model, providing all funds to foster innovation investment would maximize the output gains over the medium run.

This is not just a consequence of the fact that changes in aggregate productivity Z increase one-to-one aggregate output Y, while the elasticity of output to public capital is controlled by its efficiency χ. The reason for this is two-fold. First, since the outcomes of innovation at the firm level are uncertain, changes in innovative investment – which is what fiscal transfers stimulate – do not necessarily translate one-to-one into changes in aggregate productivity (recall here the discussion of the previous section about the elasticity of productivity to innovative investment). Second, transfers do not increase innovative investment one-to-one since the increase in the price of the research good leads to a crowding out of the innovative investment that firms undertake with their own resources.

Finally, consider the scenario where all funds are allocated to transfers to fund the accumulation of private physical capital (red lines). In this counterfactual, we observe, for both high and low public investment efficiency, that the output increase is clearly smaller than in the baseline and in the previous counterfactuals. Indeed, the effect on output during the first years is even slightly negative. The reason for this result is the combination of the following forces. First, as in the case of innovation transfers, there is a crowding out of private capital accumulation financed via private resources, which explains why transfers to private capital have smaller effects on output than when funds are allocated toward public investment. Second, the elasticity of output to private capital, α, is smaller than one, contrary to the case of aggregate productivity. Hence, everything else equal, an increase in the stock of capital has a smaller effect on output than an increase in productivity. At the same time, hours worked decline because of the positive wealth effect (recall Figure 2). This is enough to counteract the increase in private capital leading to a small fall in output.

In sum, the alternative scenarios considered in this section point towards relatively large effects of transfers that encourage innovative investment, which can be only compared to the build-up of public capital when this is highly productive. On the other hand, the gains from transfers allocated to build private physical capital are more limited than the other options considered here.

4.4 Fiscal Multipliers

In previous sections, we have explored the aggregate consequences of the current allocation of NGEU transfers, and we have investigated the effects of alternative fiscal arrangements. In this section, we summarize those results by computing cumulative fiscal multipliers, which can serve as a useful summary statistic for the effects of the NGEU program. More precisely, we define the cumulative output multiplier associated with the NGEU transfers at horizon h as:

$$\mathcal{M}_h = \frac{\sum_{j=0}^{h} \beta^j (Y_{t+j} - Y_{t-1})}{\sum_{j=0}^{h} \beta^j (T_{t+j}^{EU} - T_{t-1}^{EU})}, \tag{34}$$

where $t-1$ is the time period before NGEU transfers start to take place.

Figure 4 shows the cumulative fiscal multipliers for the baseline simulation, as well as for each of the alternative allocations of funds considered in section 4.3. As before, the left panel shows the case of high efficiency of public capital and the right panel considers the calibration with a lower efficiency of public capital.

Fiscal multipliers in the baseline simulation range between 0.80 and 0.47 in the year after the NGEU program expires. This means that, according to our simulation, for each euro of NGEU transfers output would increase on average by roughly 0.63 euros.

Comparing the fiscal multipliers associated with each of the alternative scenarios considered we observe that the main messages of the previous section hold when results are presented in these terms. The fiscal multipliers for public capital range between 1.05 (high public capital efficiency) and 0.42 (low public capital efficiency). The average fiscal multiplier associated with the counterfactual where all transfers go into innovative investment is close to the baseline with a high efficiency of public capital but substantially larger when the efficiency of public capital is low, with an average of roughly 1.1. On the contrary, the fiscal multiplier associated with transfers to private physical capital falls slightly into negative territory, in line with the effects on output described in the previous section.

The multipliers grow at longer horizons, as the benefits of the increased investment continue after the end of the NGEU expenditure. Computing a present-discounted-value version of these fiscal multipliers allows the analysis to take these long-term benefits into effect:

$$\mathcal{M}_{pdv} = \frac{\sum_{j=0}^{\infty} \beta^j (Y_{t+j} - Y_{t-1})}{\sum_{j=0}^{\infty} \beta^j (T_{t+j} - T_{t-1})}. \tag{35}$$

In this case, since we focus on the long run, the denominator includes both the transfers from the NGEU program and also the increase in fiscal expenditures required to later maintain the new stock of public capital built with the NGEU funds.

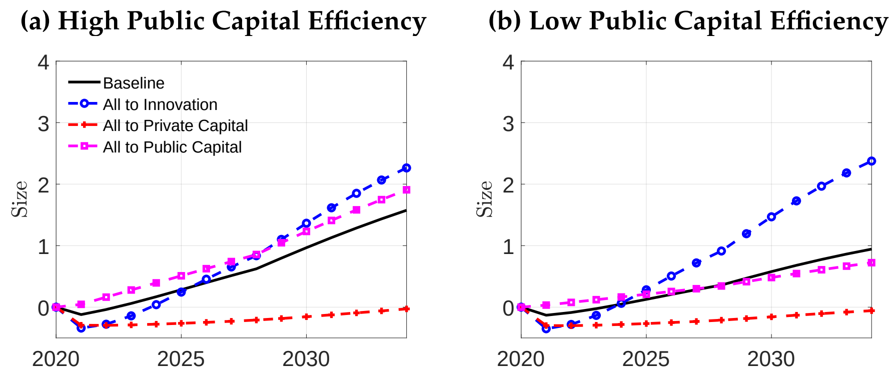

Figure 4: Fiscal Multipliers

Notes: Dynamic cumulative fiscal multipliers, computed as in equation (34). The left panel shows the effects with high efficiency of public capital, $\chi = 0.15$, and the right panel for a lower efficiency of public capital, $\chi = 0.05$. The black lines display our baseline simulation, see section 4.2. The remaining lines consider the counterfactual simulations, where all NGEU funds are allocated to either public capital (pink line, squares), innovation transfers (blue line, circles), or private capital transfers (red line, circles).

	Baseline	All to Pub. Inv.	All to Innovation	All to Priv. Inv.
High Effi. Pub. Capital	4.37	5.28	4.64	1.84
Low Effi. Pub. Capital	1.88	1.73	3.48	0.55

Table 2: Long-run Cumulative Multipliers

Notes: Discounted cumulative multipliers computed as in equation (35). First row shows the case for a high efficiency of public capital, and the second row for a low efficiency. Each column displays the results under each alternative allocation of the NGEU funds considered.

Table 2 shows the results in these terms. As can be observed, multipliers can be significantly higher than unity when the long run is included in the analysis and the fiscal expenditure is successful in increasing output.[26] Again, the table shows our previous point that increasing innovative investment tends to deliver higher returns in terms of output, unless the public stock of capital built is highly efficient.

These results showcase the fact that an increase in public expenditure or transfers, with multipliers close to unity in the short term, can have a much bigger impact in the

[26]Perhaps it could seem surprising that the long-run multipliers associated with the fiscal stimulus are meaningfully large. In Appendix B we show, following Ramey (2020), that one of the determinants of the multiplier is the steady-state ratio of public investment to GDP. Namely, if we started from a higher level of public investment, then long-run multipliers would be reduced.

long run if it is able to successfully leverage improvements in innovation and productivity. Comparing the multipliers across different levels of public investment productivity shows that a careful design of the investment package is key to achieving a high impact on output in the long run.

5 Conclusion

In this paper, we have introduced an endogenous growth model of firm dynamics based on Atkeson and Burstein (2019), extended with elastic labor supply, productivity-enhancing public capital, and fiscal transfers to fund innovation and private capital. Calibrating the model to the Spanish economy, we have used the framework to assess the aggregate effects of the different elements of the NGEU program on output and its components, including productivity.

We have found that under the current allocation of funds in the NGEU program in Spain, in the baseline calibration of our model, aggregate output growth would be boosted by 0.13 percentage points per year during the duration of the program and at the end of this period the output level would be roughly 1.1% higher, with the endogenous response of productivity to the fiscal stimulus accounting for an important share of this boost to output. Using the model as a laboratory to build counterfactuals, we have shown that transfers to innovative investment are particularly effective, only matched by increases in a stock of public capital when this is highly efficient. Although they are close to unity in the medium term, we find substantially higher multipliers in the long run, as the positive effects of these interventions outlast most of their costs. For this theoretical result to be achieved in practice, though, it is important that the design of the fiscal package actually succeeds in improving innovation and productivity.

In future work it would be interesting to extend our current framework to a multi-country model, as in Pfeiffer et al. (2022), to explore the cross-country spillovers of the NGEU funds. Since the model is particularly well suited to analyze the long-term effects of shocks and policies, it could also be used to assess the long-term consequences of changes in population growth (Jones, 2022) or be extended to assess climate change policies within an endogenous growth framework (Hassler et al., 2021). Finally, it would be of particular interest to enrich the model to assess the effects of transfers on capital and productivity when firms face financial frictions, along the lines of Moll (2014) and Garcia-Macia (2017).

Compliance with Ethical Standards

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Conflict of interest: Rubén Domínguez-Díaz declares that he has no conflict of interest. Samuel Hurtado declares that he has no conflict of interest. Carolina Menéndez declares that she has no conflict of interest.

Ethical approval: This article does not contain any studies with human participants or animals performed by any of the authors

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A Model Appendix

In this appendix we provide the details of the model outlined in the main text, including the first order conditions associated with the optimization problems of agents.

A.1 Households

The solution to the household problem outlined in the main text and described by equations (1), (2), and (3), satisfies the following optimality conditions:

$$\frac{1}{C_t/H_t} = \mathbb{E}_t \beta (1+R_t) \frac{1+\tau_t^C}{1+\tau_{t+1}^C} \frac{1}{C_{t+1}/H_{t+1}}, \tag{36}$$

$$\frac{1}{C_t/H_t} = \mathbb{E}_t \beta \frac{1+\tau_t^C}{1+\tau_{t+1}^C} \frac{1}{C_{t+1}/H_{t+1}} \frac{1}{q_t} \left(r_{t+1}^k (1-\tau_{t+1}^{\text{Corp}}) + q_{t+1} \left(1 - \delta_K - \frac{\partial \Xi^I (I_{t+1}, K_{t+1})}{\partial K_{t+1}} \right) \right), \tag{37}$$

$$\frac{1}{q_t} = 1 - \frac{\partial \Xi^I (I_t, K_t)}{\partial I_t}, \tag{38}$$

$$\begin{aligned} \kappa \left(\frac{L_t^P + L_t^R}{H_t} \right)^{\varphi} (1+\tau_t^C) \frac{C_t}{H_t} =& (1-\tau_t^L) W_t^R - \frac{\partial \Xi^L (L_t^R, L_{t-1}^R, L_t^P, L_{t-1}^P)}{\partial L_t^R} \\ & - \mathbb{E}_t \frac{1}{1+R_t} \frac{\partial (L_{t+1}^R, L_t^R, L_{t+1}^P, L_t^P)}{\partial L_t^R}, \end{aligned} \tag{39}$$

$$\kappa \left(\frac{L_t^P + L_t^R}{H_t}\right)^{\varphi} (1+\tau_t^C)\frac{C_t}{H_t} = (1-\tau_t^L)W_t^P - \frac{\partial \Xi^L(L_t^R, L_{t-1}^R, L_t^P, L_{t-1}^P)}{\partial L_t^P} \\ - \mathbb{E}_t \frac{1}{1+R_t}\frac{\partial (L_{t+1}^R, L_t^R, L_{t+1}^P, L_t^P)}{\partial L_t^P}, \tag{40}$$

Equations (36) and (37) are the Euler equations for risk-free bonds and capital, respectively. q_t above denotes Tobins's q – the marginal value of capital in terms of consumption units – and is given in equation (38). The final two equations summarize the labor supply of research and production labor of the household.

A.2 Firms

A.2.1 Final Good Producer

The solution to the final good producer's problem (4) delivers the following system of demand equations for intermediate goods:

$$y_t(z) = p_t(z)^{-\rho} Y_t, \tag{41}$$

A.2.2 Intermediate Good Producers

We divide the problem solved by intermediate good producers into two problems. The first one is a static problem where firms optimally choose production inputs and prices. The second problem is a dynamic problem where firms decide how much to invest in innovation.

Static Problem. The static problem of a firm with productivity index z consists of choosing labor $l^p(z)$, capital $k(z)$, and prices $p(z)$ to maximize variable profits (6), subject to the demand equation (41) and the production technology (5). The solution to that problem is characterized by the following equations:

$$\frac{k_t(z)}{l_t^p(z)} = \frac{\alpha}{1-\alpha}\frac{W_t^P}{r_t^k}, \tag{42}$$

$$W_t^P = \lambda_t(z)(1-\alpha)\frac{y_t(z)}{l_t^p(z)}, \tag{43}$$

$$\lambda_t(z) = \max\left\{0, \frac{1}{z}\mathrm{MC}_t\right\}, \tag{44}$$

$$p_t(z) = \mu\frac{1}{z}MC_t, \tag{45}$$

$$\mu = \min\{\frac{\rho}{\rho-1}, \bar{\mu}\}, \tag{46}$$

$$\mathrm{MC}_t \equiv \frac{1}{(K_t^G)^\chi}\left(\frac{W_t^P}{1-\alpha}\right)^{1-\alpha}\left(\frac{r_t^k}{\alpha}\right)^\alpha \tag{47}$$

Equation (42) shows that the capital-to-labor ratio chosen by intermediate good firms is independent of the idiosyncratic productivity index z, albeit the levels of these variables do not need to be. Equation (43) is the equation that defines labor demand for production work of a firm with productivity index z, where $\lambda_t(z)$ is the Lagrange multiplier as given by (44).[27]

Equation (45) states that intermediate good producers charge a constant markup μ over marginal costs MC_t/z, defined in (47). As in Atkeson and Burstein (2019), we assume that the markup is given by the minimum between the monopoly markup $\rho/\rho-1$

[27]The complementary slackness conditions is given by $\lambda_t(z)\left(y_t(z) - z(K_t^G)^\chi k_t(z)^\alpha l_t^p(z)^{1-\alpha}\right) = 0$.

and the technology gap with respect to the second most productive firm producing the same product with productivity index $z/\bar{\mu}$.

Dynamic Problem. The dynamic problem of an incumbent firm consists of choosing innovative investment to maximize:

$$V_t(z) = \max_{\tilde{x}_t^m(z), \tilde{x}_t^c(z)} \pi_t(z)(1-\tau_t^{\text{Corp}}) - P_t^r(\tilde{x}_t^m(z) + \tilde{x}_t^c(z)) + \mathbb{E}_t \frac{V_{t+1}(z')}{1+R_t}\left((1-\delta_0) + h\left(\frac{\tilde{x}_t^m(z)}{s_t(z)}\right)\right), \tag{48}$$

where variable profits $\pi_t(z)$ are defined in (6). The continuation value of the firm $V_{t+1}(z')$ is weighted by two terms. The first of them $1-\delta_0$ corresponds to the probability of keeping an existing product. The second term $h(\tilde{x}_t^m(z)/s_(z))$ is the probability of having the opportunity to invest in a new product.

We solve the problem (48) following the same steps as in Atkeson and Burstein (2019). Namely, one first can easily show that variable profits scale with $s_t(z)$, that is $\pi_t(z) = s_t(z)(1-\tau_t^{\text{Corp}})\frac{\mu-1}{\mu}Y_t$. Second, one can show that innovative investment scales with $s_t(z)$ as well – $x_t^j(z) = s_t(z)x_t^m$ for $j \in \{c, m, e\}$. This leads to $V_t(z) = V_t s_t(z)$, where V_t is given by:

$$V_t = \max_{\tilde{x}_t^m, \tilde{x}_t^c}(1-\tau_t^{\text{Corp}})\frac{\mu-1}{\mu}Y_t - P_t^r(\tilde{x}_t^m + \tilde{x}_t^c) + \mathbb{E}_t \frac{V_{t+1}}{1+R_t}\left((1-\delta_0)\zeta(x_t^c) + \eta_m h(x_t^m)\right)\frac{Z_t^{\rho-1}}{Z_{t+1}^{\rho-1}} \tag{49}$$

The first-order conditions for innovative investment for incumbent firms and the free-entry conditions for new entrants are therefore given by:

$$x_t^m: \quad P_t^r = \frac{1}{1+R_t} V_{t+1} \eta_m h'(x_t^m) \left(\frac{Z_t}{Z_{t+1}} \right)^{\rho-1} \tag{50}$$

$$x_t^c: \quad P_t^r = \frac{1-\delta_0}{1+R_t} V_{t+1} \xi'(x_t^c) \left(\frac{Z_t}{Z_{t+1}} \right)^{\rho-1} \tag{51}$$

$$\text{Free-entry}: \quad P_t^r = \frac{1}{1+R_t} V_{t+1} \eta_e \left(\frac{Z_t}{Z_{t+1}} \right)^{\rho-1} \tag{52}$$

A.2.3 Research Good Producer

The maximization problem of the research good producer (14) delivers the following first-order conditions for the demand of research labor:

$$P_t^R A_t^R \left(K_t^G \right)^{\chi} Z_t^{\phi-1} = W_t^R \tag{53}$$

A.3 Aggregation

In this section we show that aggregate output Y_t can be written as:

$$Y_t = Z_t (K_t^G)^{\chi} (K_t)^{\alpha} \left(L_t^p \right)^{1-\alpha}. \tag{54}$$

We first start by noting that using (42) and (45) the demand for intermediate goods (41) can be written as:

$$\left(K_t^G \right)^{\chi} \left(\frac{K_t}{L_t^P} \right)^{\alpha} l_t^p(z) = (\mu \mathrm{MC}_t)^{-\rho} Y_t z^{\rho-1}. \tag{55}$$

Next, using the definition of the Lagrange multiplier (44) together with the pricing condition (45), we can write the return on labor (43) as a function of aggregate output and aggregate production work:

$$W_t^p = \frac{1-\alpha}{\mu} \frac{Y_t}{L_t^p}, \tag{56}$$

and similarly for the return on private physical capital:

$$r_t^k = \frac{\alpha}{\mu} \frac{Y_t}{K_t}, \tag{57}$$

Equations (56) and (57) allows us to write marginal costs MC_t in equation (47) as:

$$\mu \text{MC}_t = \frac{Y_t}{(K_t^G)^\chi (K_t)^\alpha \left(L_t^p\right)^{1-\alpha}} \tag{58}$$

Using (58) into (55) and aggregating over z gives us:

$$\left((K_t)^\alpha \left(L_t^p\right)^{1-\alpha}\right)^{1-\rho} = Y_t^{1-\rho} \sum_z M_t(z) Z^{\rho-1} \tag{59}$$

Rearranging (59) and defining Z_t as in (7) gives us the desired result.

A.4 Balanced Growth Path

Our economy features endogenous growth. Therefore we detrend the equilibrium variables by their constant balanced-growth-path (BGP) growth rates to obtain a stationary equilibrium.

At the BGP we have that $\{Y_t, C_t, K_t, P_t^r, V_t\}$ grow at the growth rate of output g_Y. Productivity Z_t grows at g_z, given by (21). Hours worked $\{L_t^R, L_t^P\}$ grow at the same rate as population, g_H. Wages $\{W_t^R, W_t^P\}$ grow at the growth rate per capita output $g_Y - g_H$. Asset returns $\{q_t, R_t, r_t^k\}$ are already stationary. Furthermore, as in Atkeson and Burstein (2019), we restrict our attention to the BGP where innovative investment $\{x_t^e, x_t^m, x_t^c\}$, and hence production of research good Y_t^r is constant. Finally, we also let public capital K_t^G grow at the same rate as output, g_Y, to have a constant public-investment-to-output ratio at the BGP, and assume that tax rates are constant at the BGP.

We denote by small-case letters detrended variables. That is, for a variable X_t we have that $x_t \equiv X_t/\exp(tg_x)$, where g_x is the constant growth rate of X_t at the BGP, is constant at the BGP. Following this notation we can write the system of equilibrium equations in terms of stationary variables as:

$$\frac{1}{c_t} = \mathbb{E}_t \beta (1 + R_t) \frac{1 + \tau_t^C}{1 + \tau_{t+1}^C} \exp(g_Y - g_H) \frac{1}{c_{t+1}} \tag{60}$$

$$\frac{1}{c_t} = \mathbb{E}_t \exp(g_Y - g_H) \frac{1 + \tau_t^C}{1 + \tau_{t+1}^C} \frac{1}{c_{t+1}} \frac{1}{q_t} \left(r_{t+1}^k (1 - \tau_t^{Corp}) + q_{t+1} \left(1 - \delta_K - \frac{\partial \Xi^I (i_{t+1}, k_{t+1})}{\partial k_{t+1}} \right) \right) \tag{61}$$

$$\frac{1}{q_t} = 1 - \frac{\partial \Xi^I (i_t, k_t)}{\partial i_t}, \tag{62}$$

$$\begin{aligned} \kappa \left(\frac{l_t^P + l_t^R}{h_t} \right)^{\varphi} (1 + \tau_t^C) \frac{c_t}{h_t} \exp(g_Y - g_H) =& (1 - \tau_t^L) w_t^R \exp(g_Y - g_H) - \frac{\partial \Xi^L (l_t^R, l_{t-1}^R, l_t^P, l_{t-1}^P)}{\partial l_t^R} \\ & - \mathbb{E}_t \frac{1}{1 + R_t} \frac{\partial (l_{t+1}^R, l_t^R, l_{t+1}^P, l_t^P)}{\partial l_t^R}, \end{aligned} \tag{63}$$

$$\begin{aligned} \kappa \left(\frac{l_t^P + l_t^R}{h_t} \right)^{\varphi} (1 + \tau_t^C) \frac{c_t}{h_t} \exp(g_Y - g_H) =& (1 - \tau_t^L) w_t^P \exp(g_Y - g_H) - \frac{\partial \Xi^L (l_t^R, l_{t-1}^R, l_t^P, l_{t-1}^P)}{\partial l_t^P} \\ & - \mathbb{E}_t \frac{1}{1 + R_t} \frac{\partial (l_{t+1}^R, l_t^R, l_{t+1}^P, l_t^P)}{\partial l_t^P}, \end{aligned} \tag{64}$$

$$p_t^r = \frac{\exp(g_Y - (\rho-1)g_Z)}{1+R_t} v_{t+1} \eta_m h'(x_t^m) \left(\frac{z_t}{z_{t+1}}\right)^{\rho-1} \tag{65}$$

$$p_t^r = \frac{\exp(g_Y - (\rho-1)g_Z)}{1+R_t} v_{t+1} (1-\delta_0) \zeta'(x_t^c) \left(\frac{z_t}{z_{t+1}}\right)^{\rho-1} \tag{66}$$

$$p_t^r = \frac{\exp(g_Y - (\rho-1)g_Z)}{1+R_t} v_{t+1} \eta_e \left(\frac{z_t}{z_{t+1}}\right)^{\rho-1} \tag{67}$$

$$p_t^R Y_t^R = w_t^R l_t^R \tag{68}$$

$$w_t^p = \frac{1-\alpha}{\mu} \frac{y_t}{l_t^p}, \tag{69}$$

$$r_t^k = \frac{\alpha}{\mu} \frac{y_t}{k_t}, \tag{70}$$

$$\begin{aligned} v_t =& \max_{\tilde{x}_t^m, \tilde{x}_t^c} (1-\tau_t^{\text{Corp}}) \frac{\mu-1}{\mu} y_t - p_t^r (\tilde{x}_t^m + \tilde{x}_t^c) \\ &+ \mathbb{E}_t \exp(g_Y - (\rho-1)g_Z) \frac{v_{t+1}}{1+R_t} \left((1-\delta_0)\zeta(x_t^c) + \eta_m h(x_t^m)\right) \frac{z_t^{\rho-1}}{z_{t+1}^{\rho-1}} \end{aligned} \tag{71}$$

$$y_t = z_t (k_t^G)^\chi (k_t)^\alpha \left(l_t^p\right)^{1-\alpha}. \tag{72}$$

$$Y_t^r = \left(k_t^G\right)^\chi z_t^{\phi-1} l_t^r. \tag{73}$$

$$\left(\frac{\exp(g_Z)z_{t+1}}{z_t}\right)^{\rho-1} = (1-\delta_0)\zeta(x_t^c) + \eta_m h(x_t^m) + \eta_e x_t^e \tag{74}$$

$$Y_t^r = x_t^e + x_t^m + x_t^c \tag{75}$$

$$c_t + i_t + i_t^G + \Xi[i_t, k_t] + \Xi[l_t^R, l_{t-1}^R, l_t^P, l_{t-1}^P] = y_t + t_t^{EU} \tag{76}$$

B Additional simulations

In this appendix, we report several simulations that complement our main results presented in the main text of the paper. Namely, we show the robustness of our results to a lower depreciation rate of public capital and to a different value of capital adjustment costs. We then show that increasing the elasticity of productivity to innovation expenditures or increasing the starting level of the public-investment-to-GDP ratio can have consequences for multipliers. As to simplify exposition, all these previous simulations are based on the calibration with a high efficiency of public capital. Next, we discuss the crowding-out effects of the fiscal stimulus. Finally, we also present simulations under alternative spending paths of the NGEU funds.

Lower depreciation rate of public capital. Our baseline calibration assumes that the depreciation of public capital is the same as the depreciation of private capital. However, it could be that these depreciation rates are different since the composition of the stock of private and public capital differs.

We consider the robustness of this assumption by entertaining a different calibration where the depreciation rate of public capital is lower than the depreciation of private capital. More precisely, in this alternative calibration we assume that $\delta_G = \frac{2}{3}\delta_K$, following the estimates of Ramey (2020) for the US. As can be observed in Figure B.1, a lower depreciation of the capital stock has little impact on our results, with the cumulative multipliers associated with the NGEU program being very close to our baseline multipliers, observe dashed lines with square markers.

Capital adjustment costs. Our main calibration assumes a value of capital adjustment costs in line with the evidence presented in Eberly et al. (2008). However, since this evidence is based on US data it is reasonable to ask about the robustness of this choice. To address this question, we have considered an alternative calibration where capital adjustment costs are set to zero. As can be observed in Figure B.1, cumulative multipliers without adjustment costs are very in line with our main simulations, compare solid black and dashed blue lines with circles.

Higher elasticity of Z_t to innovation expenditures. As discussed in Section 4.2 our baseline simulation implies a peak-to-peak elasticity of TFP to innovation expenditures of approximately 0.11. The green dashed line with solid circle markers in Figure B.1 shows the cumulative multipliers under an alternative calibration where this elasticity is increased to 0.15. We have achieved this by increasing the productivity step of entrants, η_e, meaning that the returns in terms of productivity to innovation expenditures of entrants are now much higher. Under this alternative calibration, the macroeconomic effects of the fiscal stimulus are substantially larger, with cumulative multipliers right after the NGEU program increasing up to 1.6 (from a baseline value of 0.80). This is so because, now, the increase in innovation expenditures translates more efficiently into TFP gains, increasing the output gains of the stimulus.

(a) Cumulative multipliers

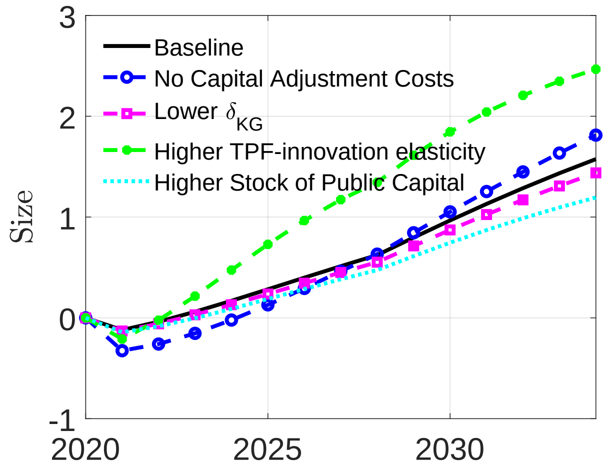

Figure B.1: Additional simulations

Notes: This figure shows the NGEU multipliers, with $\chi = 0.15$, in our baseline simulation and under alternative calibrations.

Higher stock of public capital. Ramey (2020) shows that one of the determinants of the multipliers associated with public investment is the steady-state level of the public-investment-to-GDP ratio. Namely, that paper shows that public investment multipliers tend to be larger when this ratio is below the social planner optimum. Although computing the optimal level of public investment is beyond the current paper, we still show that this logic also holds in our model. In particular, we consider an alternative calibration where the public investment-to-GDP ratio in the starting BGP is twice as large as in our baseline calibration. We report the multipliers of the NGEU program under this alternative calibration in Figure B.1, displayed with dashed light blue lines. We do observe that, effectively, multipliers would be smaller if the starting stock of public capital was higher. Namely, the cumulative multiplier would fall to 0.61 right after the NGEU program (from a baseline value of 0.80). That is, the fiscal stimulus is particularly effective when the economy departs from a starting point with low levels of public investment.

Crowding-out effects of the fiscal stimulus. One of the reasons why a fiscal stimulus may have limited effects might be that it has crowding out effects. Namely, the increase in prices generated by the fiscal stimulus may lead to a reduction of privately funded investment into physical capital or innovation. To illustrate this, we make use of the counterfactuals where all the funds are allocated either to innovation transfers or to capital transfers, as described in section 4.3.

Starting with the case where all funds are allocated to innovation expenditures, the left panel in Figure B.2 shows the path of transfers as a share of initial BGP innovation expenditures and the dynamics of total innovation expenditures. In this scenario, transfers amount to roughly 30% per year of what is spent on innovation in the initial BGP. Note, however, that total innovation expenditures increase by less, peaking at around 20%. This means that there is indeed a substantial crowding-out, even though not complete, which limits the expansionary effects of the fiscal stimulus.

A similar story holds when we look at the scenario where all funds are allocated to capital transfers, plotted in the right panel. Here, we plot transfers as a share of the initial BGP level of investment and the dynamics of total private investment in physical capital. Since the BGP investment in physical capital is much larger than innovation expenditures, now transfers represent a smaller share when plotted against BGP investment. Yet, we still observe a similar crowding out effect, with transfers representing 3% but total investment peaking at around 1%, which partly explains the small effects of capital transfers on GDP reported in the main text.

Alternative spending paths. Our baseline spending path for NGEU spending considers a uniform disbursement of funds, as described in Section 4.1. In this section, we evaluate the consequences for output of alternative paths for the disbursement of funds, which could be of interest given potential delays in the implementation of the NGEU program.

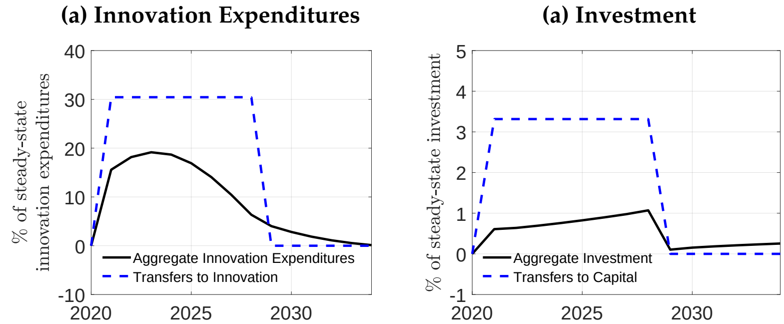

Figure B.2: Crowding-out of fiscal stimulus

Notes: Left panel: considers the simulation where all NGEU funds are allocated to innovation transfers. It shows transfers and total innovation expenditures as a share of initial steady-state innovation expenditures. Right panel: considers the simulation where all NGEU funds are allocated to capital transfers. It shows transfers and total investment in physical capital as a share of initial steady-state private capital.

We consider two alternative paths for public investment, the largest component of the NGEU funds according to our baseline. Namely, we entertain a decreasing path for public investment, as depicted in panel (a) of Figure B.3, and an increasing path for public investment, shown in panel (c).

In the medium run, the total effects on aggregate output of the different spending paths are quite similar, compare panels (b) and (d) of Figure B.3. This should not come as a surprise given that in all scenarios we maintain fixed the overall size of the program, as well as the allocation of funds to the different fiscal instruments. Indeed, the average effect on GDP growth (not shown here) remains the same under these alternative paths as under our baseline flat spending profile. The timing of the stimulus differs across the different spending paths, however. A decreasing path for public investment (bottom row, Figure B.3) implies an output effect that is more front-loaded than in our baseline, while the stimulus to output under the increasing profile (top row, Figure B.3) is more back-loaded.

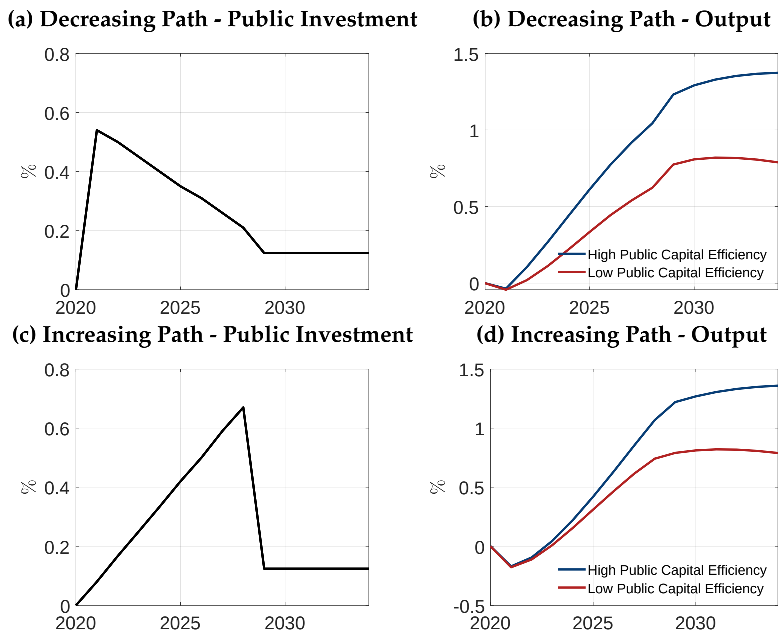

Figure B.3: Alternative Spending Paths

Notes: This figure shows the output effects under alternative paths for public investment. The top row considers a decreasing path for public investment (panel a) and the corresponding effects on output (panel b). The bottom row considers a decreasing path for public investment (panel c), and panel (d) shows the corresponding path for output.

Data Availability

The data used in this article primarily concerns those employed in the calibration exercise. These data can be accessed via the National Statistical Office of Spain (*Instituto Nacional de Estadística, INE*) and via the Central Balance Sheet Data Office (*Central de Balances*), maintained by Banco de España. The links for these databases are `https://ine.es/` and `https://www.bde.es/wbe/es/areas-actuacion/central-balances/`, respectively.